

DIGITAL FORENSICS

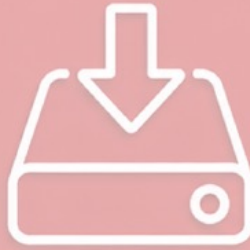
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References

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 - Bill Nelson, et al. Cengage, 2025
- Digital Forensics and Incident Response, 4th Edition,
 - Gerard Johansen, Packt, 2025
- Roadmap of digital forensics investigation process with discovery of tools, Cyber Security and Digital Forensics,
 - Anita Patil, et al. (2022):241-269.
- SWGDE 18-F-001-2.0 / SWGDE 18-F-002-2.0 /
- SWGDE 17-F-002-2.1 / SWGDE 23-Q-001-1.1 /
- NIST SP 800-61 Rev. 3
- IETF RFC 3227
- <https://www.sans.org/blog>

DAY 3

**EXAMINATION AND ANALYSIS OF THE
DIGITAL EVIDENCE**



1

Acquisition



2

**Examination
and Analysis**



3

Reporting

Privacy

- Respect the privacy rules and guidelines of your company and your legal jurisdiction.
- No information collected along with the evidence you are searching for is available to anyone who would not normally have access to this information.
 - includes access to log files (which may reveal patterns of user behavior) as well as personal data files.
- Do not intrude on people's privacy without strong justification.

Timing and Timeline Analysis

- Timeline analysis helps reconstruct the sequence of events, identify when an incident started, and correlate activities across logs, files, memory, and network evidence.
- Unsynchronized systems may record the same event with different timestamps because their clocks are incorrect or not using NTP.
- **Ensure that timestamps** from different systems are correctly interpreted and normalized.

**SEARCH FOR SPECIFIC
TERMS OR HASHES**

Searching For Documents

- Once you've got images, it's time to begin processing the evidence.
- How we proceed typically depends on the type of matter we're working,
- If we're searching for a specific files, we'll typically load the forensic image into one of our digital forensic programs to index;
 - Autopsy, Access Data's Forensic ToolKit (FTK)
- FTK allows us to conduct a search of specific terms without having to run a long search of the drive which might take hours.

Searching For Documents

- Autopsy Hash Database Lookup Module
 - calculates MD5 hash values for files
 - and looks up hash values in a database to determine if the file is known bad, known (in general), or unknown.

AUTORUNS

Autostarts

- Autostarts is the term I use to refer to software that runs automatically without being intentionally started by a user.
 - This type of software includes drivers and services that start when the computer is booted; applications, utilities, and shell extensions that start when a user logs on; and browser extensions that load when Internet Explorer is started.
 - Over 200 locations in the file system and registry allow autostarts to be configured on x64 of Windows. These locations are often referred to as Autostart Extensibility Points (ASEPs).

Autostarts

- **Malware** is software.
- Software **has** to be told when **to execute**.
- ASEP's are a very common way for both legitimate and malicious software to start on a Windows host.
- The **mechanisms** used by malware to stay on a system are commonly referred to as "**Persistence**".
- We can often find the mechanisms malware is using to stay on a system by analyzing the data collected from using the **autorunsc** tool.

Autostarts

- “**autoruns**” is a term used to refer to the collection of locations Windows looks for processes to start during boot time.
- Until dead box autoruns is improved it is recommended to gather the autoruns data from a live host when the disk and memory images are collected.
- One method of doing this is to include the Autorunsc program on the incident USB drive.

PROGRAM EXECUTION

Program Execution

- Program execution analysis **uncovers** what has run on a system
– to reveal user activities performed on the system.
- By analyzing program execution artifacts, **forensic investigators can know** what programs were installed on the system and their execution history, even the evidence of deleted programs.
- The artifacts of program executions are created and stored at various locations in hard disk, which are further utilized by OS for providing improved user experience.

Program Execution

- KAPE can collect:
 - Prefetch
 - ShimCache
 - Amcache.hve

PREFETCH

Program Execution – Prefetch

- The main purpose of prefetch files from the Windows operating system's viewpoint
 - is to **cache** pieces of information over an executable after it is run.
 - to allow the operating system **to speed up** the loading of the application in subsequent executions.
- This means that there may be **multiple prefetch files for the same application** but correspond to separate directory locations.
- The default naming convention of a prefetch file is
 - <Name_Of_Executable>-<Hash_of_Directory>.pf.
- For example, the program zoom.exe could appear as:
 - ZOOM.EXE-87652BD0.pf,
 - where 87652BD0 is a hash of the path from which zoom.exe was executed

Program Execution – Prefetch

- The **timestamps** of a prefetch file are the most beneficial aspect of the file itself.
 - The **creation date** of a prefetch file is the first time that executable was ran on a system,
 - while the **modification date** of a prefetch file is the last time that executable was ran on a system See figure below.

Name	Date created	Date modified
 ZOOM_CM_FTOIGFK88Z9VVRZO4_M-F-B8542E79.pf	8/16/2022 2:01 PM	8/16/2022 2:01 PM
 ZOOM.EXE-87652BD0.pf	8/18/2022 1:48 PM	8/18/2022 2:34 PM

Program Execution – Prefetch

Prefetch Artifacts

Prefetch Artifacts:

- Executable File Name - Process/application name.
- Executable File Path - Process/application path.
- Run Count - Indicates the number of times the process/application was run.
- Prefetch File Creation Time - File creation time.
- Prefetch File Last Modification Time - File last modification time.
- Last Run Time - Date and time when the process/application was last run.
- 2nd Recent Run Time - 2nd recent run time.
- 3rd Recent Run Time - 3rd recent run time.
- 4th Recent Run Time - 4th recent run time.
- 5th Recent Run Time - 5th recent run time.
- 6th Recent Run Time - 6th recent run time.
- 7th Recent Run Time - 7th recent run time.
- 8th Recent Run Time - 8th recent run time.
- Loaded Resources - Loaded resources.

Program Execution – Prefetch

Prefetch Artifacts

Location of Prefetch Artifacts:

- **Prefetch** artifacts located at:

```
C:\Windows\Prefetch
```

Program Execution – Prefetch

Tools

- PECmd.exe
- KAPE (runs PECmd.exe when you enable "!EZParser")

Program Execution – Prefetch

Tools

```
Created on: 2023-10-09 16:27:22
Modified on: 2023-10-09 16:38:34
Last accessed on: 2023-10-09 16:41:57

Executable name: EVIL.EXE.EXE
Hash: 26DD15B4
File size (bytes): 53,262
Version: Windows 10 or Windows 11

Run count: 3
Last run: 2023-10-09 16:38:22
Other run times: 2023-10-09 16:35:38, 2023-10-09 16:27:12

Volume information:
#0: Name: \VOLUME{01d6b97080de36cc-2c80f17b} Serial: 2C80F17B Created: 2020-11-13 03:53:10

Directories referenced: 10
00: \VOLUME{01d6b97080de36cc-2c80f17b}\USERS
01: \VOLUME{01d6b97080de36cc-2c80f17b}\USERS\
02: \VOLUME{01d6b97080de36cc-2c80f17b}\USERS\ \DESKTOP
03: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS
04: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\GLOBALIZATION
05: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\GLOBALIZATION\SORTING
06: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\REGISTRATION
07: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\SYSTEM32
08: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\SYSTEM32\EN-US
09: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\WINSXS\AMD64_MICROSOFT.WINDOWS.COMMON-CONTRC

Files referenced: 74
00: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\SYSTEM32\NTDLL.DLL
01: \VOLUME{01d6b97080de36cc-2c80f17b}\USERS\ \DESKTOP\EVIL.EXE.EXE (Executable: 1
02: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\SYSTEM32\KERNEL32.DLL
03: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\SYSTEM32\KERNELBASE.DLL
04: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\SYSTEM32\LOCALE.NLS
05: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\SYSTEM32\SHELL32.DLL
06: \VOLUME{01d6b97080de36cc-2c80f17b}\WINDOWS\SYSTEM32\USER32.DLL
```

SHIMCACHE

Program Execution – ShimCache

- The main purpose of **ShimCache** (also known as AppCompatCache) is to provide compatibility for old applications.
 - Meaning it allows users to run older software in newer Windows systems.
 - Executables that are visible in Windows Explorer are added to ShimCache for shimming the file when required, providing a better experience to the users.

Program Execution – ShimCache

Forensics Value

- ShimCache records the executable file name, file path, and the last modification date and time.
 - By analyzing the entries, we can identify whether an executable was run on a system.
- ShimCache entries are written into the hard drive when the system reboots or shuts down.
 - This feature makes Anti-Forensics (data wiping of the Registry entries) complicated. Thus, analyzing ShimCache can provide valuable information, especially, during a malware incident analysis.

Program Execution – ShimCache

ShimCache.hve Artifacts

- Path - File path.
- Last Modification Date - Last modification date.
- Last Update Date - Last update date.
- File Size - File size.
- Entry Order - Entry order in the ShimCache.
- Is Executed - Is the file executed
- Located At - File location details.

Program Execution – ShimCache

ShimCache.hve Artifacts

Location of ShimCache.hve Artifacts:

- ShimCache.hve file is located at:

```
C:\Windows\System32\config\SYSTEM
```

- The Registry Key is located at:

```
HKLM\SYSTEM\CurrentControlSet\Control\SessionManager\AppCompatCache\AppCompatCache
```

Program Execution – ShimCache

Tools

- AppCompatCacheParser.exe
- KAPE (runs AppCompatCacheParser.exe when you enable "!EZParser")

Program Execution – ShimCache

Tools

```
AppCompatCacheParser.exe -f C:\Windows\System32\config\SYSTEM --csv E:\AppCompatCache
```

C	D
Path	LastModifiedTimeUTC
C:\Users\darkcybe\AppData\Local\Temp\44DDA91F-5918-45B9-A0E9-60C9FDC872FF\dismhost.exe	6/5/2021 12:05
C:\Windows\SoftwareDistribution\Download\Install\AM_Delta_Patch_1.375.1451.0.exe	10/4/2022 22:45
C:\Windows\system32\runonce.exe	9/19/2022 6:00
0000000901a54e6602ad0000000000000000000008664Microsoft.Windows.Client.WebExperiencecw5n1h2txyewy	
C:\Program Files\WindowsApps\Microsoft.Windows.Client.WebExperience_421.20070.685.0_x64__cw5n1h2txyewy\Dashboard\Widgets.exe	10/4/2022 4:46
0000000900000050cdea0000000a000047ba00008664Microsoft.Todos8wekyb3d8bbwe	
000000090001000f0a990000000a000055f000008664Microsoft.WindowsTerminal8wekyb3d8bbwe	
C:\Program Files\WindowsApps\Microsoft.MicrosoftOfficeHub_18.2209.1071.0_x64__8wekyb3d8bbwe\Application	10/4/2022 4:47
00000009001208a1042f0000000000000000000008664Microsoft.MicrosoftOfficeHub8wekyb3d8bbwe	
00000009001208a1042f0000000a00004a6100008664Microsoft.MicrosoftOfficeHub8wekyb3d8bbwe	
00000009000b08a000010000000a000055f000008664Microsoft.WindowsCalculator8wekyb3d8bbwe	
000000090001000000640000000a000055f000008664Microsoft.WindowsAlarms8wekyb3d8bbwe	
0000000900010000002f0000000a000055f000008664Microsoft.WindowsMaps8wekyb3d8bbwe	
00000009000b08a000190000000a000055f000008664Microsoft.WindowsNotepad8wekyb3d8bbwe	
0000000901a54e6602ad0000000a00007fff00008664Microsoft.Windows.Client.WebExperiencecw5n1h2txyewy	
0000000900010000003e0000000a000055f000008664Microsoft.WindowsSoundRecorder8wekyb3d8bbwe	
00000009000b08a000060000000a000055f000008664Microsoft.Paint8wekyb3d8bbwe	
000000090005033623c90000000a000055f000008664Microsoft.XboxGamingOverlay8wekyb3d8bbwe	
0000000956bf057900090000000a000055f000008664Microsoft.WindowsStore8wekyb3d8bbwe	

AMCACHE.HVE

Program Execution – Amcache.hve

- The Amcache.hve file contains information on the executables that were executed on the system.
- The artifacts in this file can serve as a huge aid in an investigation,
 - **it records the processes recently run on the system and lists the paths of the files executed.**

Program Execution – Amcache.hve

- AmCache artifacts are important to investigations where the tracing of external storage devices, portable programs and anti-forensic programs might be required.
- The data contained in the file includes:
 - the execution paths, installation, execution, deletion times and other metadata.
 - It also stores the SHA1 hashes of the programs which can be used to compare against the hashes of malicious programs available in public database.

Program Execution – Amcache.hve

AmCache **Files** Artifacts

The artifact contains information on files used by executable programs:

- File Name - The name of the file.
- File Description - The file description.
- File Extension - File extension.
- Full Path - The full path to the file.
- File Size - The file size.
- Program Name - The name of the program associated with the file.
- Program ID – ID of the program associated with the file.
- Program Version - Program version.
- PE Header Size – Portable executable header size.
- PE Header Checksum - Portable executable header checksum.
- PE Header Hash - Portable executable header hash.
- SHA-1 Hash - SHA-1 Hash.
- Switchback Context - Switchback context.
- Language Code - Language code of the file.
- Linker Date/Time - Linker date and time of the file.
- Last Modified Date/Time - Date and time when the file was last modified.
- Last Modified Date/Time 2 - Last Modified Date/Time 2.
- Key Last Update Date/Time - Registry key last update date/time.
- Created Date/Time – Date and time when the file was created.

Program Execution – Amcache.hve

AmCache **Programs** Artifacts

The artifact contains information on programs that have been executed on the system.

- Program Name - The name of the program.
- Program Version - Program version.
- Publisher Name – Program publisher name.
- Package GUID - Package GUID.
- Product GUID - Product GUID.
- File List - List of file entries in package.
- Path List - List of paths to files.
- Install Source - Source of program installation.
- Uninstall Registry Key – Path to uninstall registry key.
- Msi Package - Msi package.
- Msi Product - Msi product.
- Language Code - Language code of the program.
- Install Date/Time - Install date/time.
- Uninstall Date/Time - Uninstall date/time.
- Key Last Update Date/Time - Registry key last update date/time.

Program Execution – Amcache.hve

AmCache Artifacts

Location of AmCache Artifacts:

- AmCache.hve file is located at:

```
C:\Windows\appcompat\Programs\Amcache.hve
```

Program Execution – **Amcache.hve**

Tools

- amcacheparser.exe
- KAPE (runs amcacheparser.exe when you enable "!EZParser")

Program Execution – Amcache.hve

Tools

```
amcacheparser.exe -r -f Amcache.hve --csv D:\<dir>
```

Timestamp	Path	Publisher	Version	SHA1
=	ABC	ABC	ABC	ABC
2023-10-15 19:27:51	c:\program files\common files\vmware\drivers\vss\comreg.exe			4ab1726215e92af6d7d8da3b7865806604a19f50
2023-10-15 19:28:04	c:\program files\dotnet\sdk\7.0.402\containers\containerize\containerize.exe	microsoft corporation	7.4.223.48041	3ac398095a4cfafaffe3b288feb2617012d270
2023-10-15 19:28:04	c:\program files\dotnet\sdk\7.0.402\containerize.exe	microsoft corporation	7.4.223.48041	3ac398095a4cfafaffe3b288feb2617012d270
2023-10-15 19:28:08	c:\program files(x86)\microsoft\edge\application\86.0.622.38\cookie_exporter.exe	microsoft corporation	86.0.622.38	79489ec6e7a8726fda3c6673c3acd5fda79e6f32
2023-10-15 19:27:51	c:\program files\dotnet\shared\microsoft.netcore.app\7.0.12\createdump.exe	microsoft corporation	7,0,1223,47720 @commit: 4a824ef37caa51072221584c64cbf15455f406ca	19bd817ac0f4c0a1633b9dc4328e412ced7bf724
2023-10-15 19:28:11	c:\program files(x86)\screenconnect\bin\roslyn\csc.exe	microsoft corporation	4.100.22.11411	b2b65fa194426a7d6a5b923813db7d189377b735
2023-10-15 19:28:11	c:\program files(x86)\screenconnect\bin\roslyn\csi.exe	microsoft corporation	4.100.22.11411	41aec63a5b5293f276081312a652c7a4c52759b6
2023-10-15 18:54:05	c:\windows\system32\csrss.exe	microsoft corporation	10.0.20348.1 (winbuild.160101.0800)	7821fb14835083bae87a66ce96b67b261bea6b2b

Program Execution – Amcache.hve

- AmCache artifacts are important to investigations where the tracing of external storage devices, portable programs and anti-forensic programs might be required.
- The data contained in the file includes:
 - the execution paths, installation, execution, deletion times and other metadata.
 - It also stores the SHA1 hashes of the programs which can be used to compare against the hashes of malicious programs available in public database.

USER ACCOUNTS

User Accounts

- NTDS.dit is the Active Directory database that stores domain information, including user accounts, groups, computer objects, and password hashes. It is a critical forensic artifact for investigating authentication, privilege escalation, account compromise, and domain-wide activity.

User Accounts

- Windows stores user accounts and security descriptors for users on the local computer in a file called SAM (Security Account Manager).
- SAM is a part of a system defined database where configuration data is stored and retrieved.
- User Accounts data can be extracted from this file.
- SAM for local users
- NTDS.dit for Active Directory

User Accounts

- User Accounts artifact helps investigators identify the users on a Windows system.
 - username, full name, level of privilege the user has, last time the system was logged into with the account, last time a failed password login attempt was made, and other information.
- Analyzing this artifact helps tie a user to any activity on the computer.
 - for example, if a file download appeared in the system and the action needs to be traced, this artifact can provide clarity about the timeline by showing which account was logged into right before the incident.
 - If the account is password protected, it can show what privileges the account holds as well.

User Accounts –

User Accounts Artifact

The artifact contains information on the user accounts in the system

- Username - Account username.
- Full Name - Full name of the account user.
- User Comment - User account comment.
- Profile Path - The path to the user profile folder.
- Account Type - User account type. It could be a default admin user, custom limited or guest account.
- Account Status - Current status of the user account. Is it "Password Required and Active", "Account Not Active", "Password not Required", or "Account Disabled and Password Not Required".
- Login Count - The number of times the user has logged into the system.
- Last Login Date/Time - Date/Time when a user last logged into the system using the account.
- Last Password Reset Date/Time - Date/Time when the password was last reset.
- Last Password Failed Date/Time - Date/Time of last incorrect password login attempt.
- Logon Script - Any logon scripts that get run when logging in as that user.
- Last Write Date/Time - Last date/time the registry key containing the artifact data was written to.

User Accounts –

User Accounts Artifacts

Location of User Accounts Artifacts:

- User accounts artifact source file is located at:

```
C:\Windows\config\SAM
```

```
C:\Windows\NTDS\NTDS.dit
```

- Within the SAM hive, the artifact data can be found at:

```
SAM\Domains\Accounts\Users
```

EMAIL

Emails – Microsoft Outlook

- Microsoft Outlook is an email application that provides:
 - personal information management, task managing, contact managing, email client, calendaring, note-taking, and web browsing.
- Microsoft Outlook was developed by Microsoft Corporation. Outlook is available for desktop (Windows and macOS) and mobile devices (Android, iOS, and Windows).

Emails – Microsoft Outlook – Forensics Value

- Mailboxes make an essential part of our lives since it is considered one of the most important methods of communication.
- The forensics of mailboxes is a crucial part of digital forensics.
- Forensic searches are carried out to investigate and find any leads of a felony or wrong acts which helps in solving a case or problem.

Emails – Microsoft Outlook – Outlook MSG Artifact

This artifact contains the .msg file information that is intended for single message objects, such as an email, an appointment, a contact, and a task. The details you can view include:

- Attachments - Attachments.
- BCC - BCC email.
- CC - CC email.
- Body - Body.
- Headers - Headers.
- Importance - It indicates the message's importance.
- Message ID - Message ID.
- Sender - Sender.
- Subject - Subject.
- To - Email to.
- Message Date - The date and time of the message.
- Message Last Modification - The date and time the message was last modified.
- Message Creation Date - The date and time the message was created.

Emails – Microsoft Outlook – Outlook Email Artifacts

Outlook stores email artifacts at the following location:

```
c:\Users\%username%\Appdata\Local\Microsoft\
Outlook\*.pst
```

MEMORY FORENSICS

Memory Forensics

- **Memory forensics** is the process of capturing and analyzing a computer's memory (RAM) to uncover valuable digital artifacts.
- These artifacts can include:
 - information about currently running processes, open files, network connections, system and user configurations, and,
 - evidence of malware or hacker activity that might not be detectable through traditional disk-based analysis.
 - This is because many advanced threats operate solely in memory, leaving little to no footprint on the hard drive.

Memory Forensics –

Forensics Value of Memory (RAM)

- **Running Processes:**
 - Details about processes currently running on the system
- **Network Connections:**
 - Information about active and listening network connections
 - Can help trace back network-based attacks or data exfiltration attempts.
- **Binary Executables and Loaded DLLs:**
 - identifying unauthorized or malicious code execution
- **Registry Keys and Configuration Settings:**
 - Real-time data on system configuration and registry settings, which can indicate system tampering or policy violations.
- **Digital Artifacts:**
 - User actions, such as command history, file access, and clipboard contents, can be pivotal in reconstructing user behavior or attacker activities.

Memory Forensics – Tools

- **Volatility** is an open-source memory forensics framework used in Malware analysis and Incident Response.
- This framework is CLI-based and is programmed in Python. It's supported on Windows, Linux, and MacOS.

Memory Forensics – Tools

```
remnux@remnux:~/Desktop/Memory$ vol3 -f MemoryDump.mem windows.pslist
Volatility 3 Framework 2.3.0
Progress: 100.00
PDB scanning finished
```

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime
4	0	System	0xad8185883180 157	-	N/A	False	2023-05-21	22:27:10.000000	N/A
108	4	Registry	0xad81858f2080	4	-	N/A	False	2023-05-21 22:26:54.000000	
332	4	smss.exe	0xad81860dc040	2	-	N/A	False	2023-05-21 22:27:10.000000	
452	444	csrss.exe	0xad81861cd080	12	-	0	False	2023-05-21 22:27:22.000000	
528	520	csrss.exe	0xad8186f1b140	14	-	1	False	2023-05-21 22:27:25.000000	
552	444	wininit.exe	0xad8186f2b080	1	-	0	False	2023-05-21 22:27:25.000000	
588	520	winlogon.exe	0xad8186f450c0	5	-	1	False	2023-05-21 22:27:25.000000	
676	552	services.exe	0xad8186f4d080	7	-	0	False	2023-05-21 22:27:29.000000	
696	552	lsass.exe	0xad8186fc6080	10	-	0	False	2023-05-21 22:27:29.000000	
824	676	svchost.exe	0xad818761d240	22	-	0	False	2023-05-21 22:27:32.000000	
852	552	fontdrvhost.exe	0xad818761b0c0	5	-	0	False	2023-05-21 22:27:33.000000	
860	588	fontdrvhost.exe	0xad818761f140	5	-	1	False	2023-05-21 22:27:33.000000	
952	676	svchost.exe	0xad81876802c0	12	-	0	False	2023-05-21 22:27:36.000000	
1016	588	dwm.exe	0xad81876e4340 15	-	1	False	2023-05-21	22:27:38.000000	N/A
448	676	svchost.exe	0xad8187721240	54	-	0	False	2023-05-21 22:27:41.000000	
752	676	svchost.exe	0xad8187758280	21	-	0	False	2023-05-21 22:27:43.000000	

Memory Forensics – Tools

```
remnux@remnux:~/Desktop/Memory$ vol3 -f MemoryDump.mem windows.malfind
Volatility 3 Framework 2.3.0
Progress: 100.00 PDB scanning finished
PID Process Start VPN End VPN Tag Protection CommitCharge PrivateMemory
5896 oneetx.exe 0x400000 0x437fff VadS PAGE_EXECUTE_READWRITE 56
4d 5a 90 00 03 00 00 00 MZ.....
04 00 00 00 ff ff 00 00 .....
b8 00 00 00 00 00 00 00 .....
40 00 00 00 00 00 00 00 @.....
00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 .....
00 00 00 00 00 01 00 00 .....
0x400000: dec ebp
0x400001: pop edx
0x400002: nop
0x400003: add byte ptr [ebx], al
0x400005: add byte ptr [eax], al
0x400007: add byte ptr [eax + eax], al
0x40000a: add byte ptr [eax], al
7540 smartscreen.ex 0x2505c14000 0x2505c15ffff VadS PAGE_EXECUTE_READWRITE 1
```


Memory Forensics – Tools

```
vol3 -f MemDump.mem windows.netscan
```

0xdb898221dae0	UDPv4	0.0.0.0	4500	*	0	984	svchost.exe	2021-05-10	10:18:24.000000
0xdb8982313390	TCPv4	0.0.0.0	49667	0.0.0.0	0	LISTENING	1552	spoolsv.exe	2021-05-10 10:18:24.000000
0xdb8982313390	TCPv6	::	49667	::	0	LISTENING	1552	spoolsv.exe	2021-05-10 10:18:24.000000
0xdb89825e68f0	TCPv4	192.168.233.128	49728	203.99.187.137	443	CLOSED	6336	smsfwdr.exe	2022-01-17 09:04:55.000000
0xdb89827e33b0	UDPv4	0.0.0.0	5353	*	0	1272	svchost.exe	2021-05-10	12:01:40.000000
0xdb8982aa2730	TCPv4	0.0.0.0	49676	0.0.0.0	0	LISTENING	576	lsass.exe	2021-05-10 10:18:33.000000
0xdb8982aa2730	TCPv6	::	49676	::	0	LISTENING	576	lsass.exe	2021-05-10 10:18:33.000000
0xdb8982abc650	UDPv4	0.0.0.0	5355	*	0	1272	svchost.exe	2021-05-10	12:01:40.000000
0xdb8982abc650	UDPv6	::	5355	*	0	1272	svchost.exe	2021-05-10	12:01:40.000000
0xdb8982b23860	TCPv4	0.0.0.0	49676	0.0.0.0	0	LISTENING	576	lsass.exe	2021-05-10 10:18:33.000000
0xdb898367c6d0	TCPv4	192.168.233.128	49732	144.76.62.10	8080	SYN SENT	6336	smsfwdr.exe	2022-01-17 09:06:37.000000
0xdb898369fec0	TCPv4	192.168.80.129	139	0.0.0.0	0	LISTENING	4	System	2021-05-10 12:01:12.000000
0xdb8983700960	UDPv4	0.0.0.0	5353	*	0	1272	svchost.exe	2021-05-10	12:01:40.000000
0xdb8983700960	UDPv6	::	5353	*	0	1272	svchost.exe	2021-05-10	12:01:40.000000
0xdb898371e1a0	UDPv4	0.0.0.0	0	*	0	1272	svchost.exe	2022-01-17	09:06:25.000000
0xdb898371e1a0	UDPv6	::	0	*	0	1272	svchost.exe	2022-01-17	09:06:25.000000
0xdb89837cfd00	TCPv4	192.168.233.128	49727	190.117.206.153	443	CLOSED	6336	smsfwdr.exe	2022-01-17 09:04:30.000000

Lab 3: Examination and Analysis of The Digital Evidence

Part 1